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Lesson Notes

MODULE 6 | LESSON 1

GOING DEEPER WITH BAYESIAN NETS

Reading Time	30 minutes
Prior Knowledge	Binning
Keywords	Degree distribution, Complement of a cumulative distribution function (CCDF), Network topology, Components, Communities, Clusters, Power law

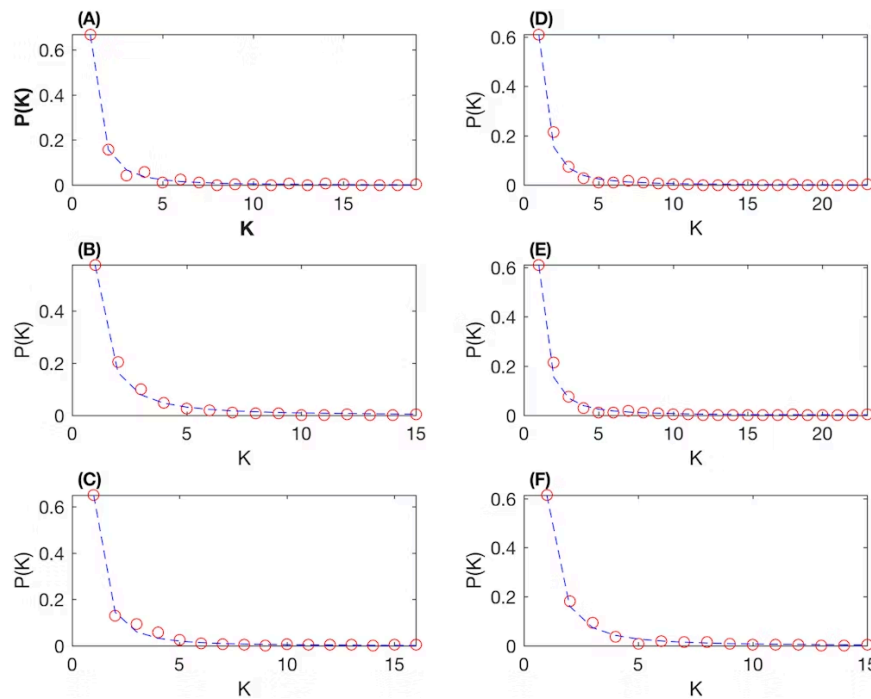
In the last module, we started working with very simple Bayesian networks, in both theory and (Python) application. In this module, we continue our experimentation but with larger Bayesian networks, including (in Lesson 3) discussion of one of the classic examples, well known for its complexity. However, in this lesson, we mostly learn more theory and vocabulary related to these powerful tools, building up our intuition for the later lessons in this module.

1. Degree Distributions

The most interesting thing about Lesson 3 of this module is that in providing the foundational insights into how to describe and analyze a network, it coincidentally provides the previously missing context for a lot of the information we have already seen in the Portfolio Management course.

You have already seen figure 1, and if you had taken a moment during Portfolio Management to interpret the graphs, you likely already made sense of them: These are just histograms showing the relative frequency of degree numbers for different networks. With the reading by Coscia, however, you learn how typical the shapes of these distributions are. In fact, these shapes strongly resemble the shape in Coscia's example of the protein-protein interaction network displayed in Figure 6.9(a).

Figure 1: Degree Distribution



Source: Guo, Xue et al. "Development of Stock Correlation Networks using Mutual Information and Financial Big Data." *PLOS One*, 2018, <https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0195941>.

2. Topologies

2.1 Average Degree

The figure below is aimed at refreshing your memory about the table summarizing the basic properties of the networks referenced above. (In fact, the "Degree Distribution" columns reduce the frequency information into three bins.) In addition, the "AD" column from the "Topologies" side of the table tells us the average degree. Coscia mentions average degree as a potentially attractive summary statistic for a network but then reminds us that it is ultimately insufficient given that very, very different networks can share this number (97). The average degree (like any mean) can be misleading, as we see when we compare these numbers, all in the range [8, 10], to their distribution: The mean is not the mode, or even close to it; most nodes have only two connections.

Figure 2: Table of Degree Distribution, Average and Longest Distance, and Power Law Exponent

Network	Degree Distribution			Topologies		
	1	2-6	≥ 7	AD	LD	PW
MI	187(66.79%)	82(29.29%)	11(3.93%)	9.14194	23(601058-601099)	2.08
MI _{s1}	161(57.50%)	110(39.29%)	9(3.21%)	8.79831	24(600568-600705)	1.82
MI _{s2}	182(65.00%)	88(31.43%)	10(3.57%)	9.6154	27(600315-601398)	2.17
CC	171(61.07%)	95(33.93%)	13(4.64%)	8.45125	31(600688-600528)	1.98
CC _{s1}	175(62.50%)	92(32.86%)	13(4.64%)	8.45125	31(600528-601998)	1.98
CC _{s2}	172(61.43%)	94(33.57%)	14(5.00%)	8.65868	26(600568-601939)	1.93

AD: average distance; LD: longest distance; PW: power law exponent; MI, MI_{s1}, MI_{s2}: network using mutual information and the whole dataset(Fig 4), data of stage 1 (S2 Fig) and data of stage 2 (S3 Fig), respectively; CC, CC_{s1}, CC_{s2}: networks using the correlation coefficient and the whole dataset, data(Fig 5) of stage 1 (S4 Fig) and data of stage 2 (S5 Fig), respectively.

<https://doi.org/10.1371/journal.pone.0195941.t001>

Source: Guo, Xue et al. "Development of Stock Correlation Networks using Mutual Information and Financial Big Data." *PLOS One*, 2018, <https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0195941>.

2.2 Power Law

There is at least a third property of networks that Coscia shares, which we can also apply to our interpretation of the table in figure 2. The last column of the table, “PW,” indicates the power law exponent. All of these are near two, with four of the six below two. This is consistent with Coscia’s generalization that, for “real world” networks whose degree distribution follows a power law, the scaling factor $1/\alpha$ tends to be close to two (102). It is hoped that you enjoy reading about the scale-invariant or *fractal* nature of many networks such as these in this part of the Coscia reading. Recall from Coscia’s earlier definition that the power law describes the relationship between two quantities where a relative change in one corresponds to a proportional relative change in the other (so the second variable changes as a power of the first)” (34). Again we learn in this module more about the concept of the power law (as it relates to degree distribution), and we can apply this knowledge retroactively to example networks with which we are already familiar.

2.3 Components vs. Communities and Clusters

In this lesson’s main reading, you will also learn about the connected components of a network, or “subgraphs whose nodes can be reached from one other by following the edges of the network” (Coscia 115). You will also learn that these components can be strong or weak and in or out. “Community detection” is also introduced, as the network analysis equivalent of what we already know by its machine learning term, “clustering” (Coscia 136). While some authors treat components and communities as interchangeable synonyms (Meghanathan 3), Coscia does *not*. Pay attention to the definitional differences. In essence, a community suggests some additional, possibly deeper commonness among its nodes than a single connection that holds a component together. According to Meghanathan, a “community in a network graph is a subset of the vertices that have a relatively larger density of edges among themselves than to the rest of the vertices” (1).

With this understanding of community (as a special subset of components), we can revisit several additional graphs from the last course and make even more sense of them or at least give them some richer context. Take for example figure 1 from Scaramozzino et al. (8), which shows a directed graph of causality links as connections between stocks. It does not show the entire network of links, which would indicate the components of the network, but rather the *subset* of connections where the links have Z scores greater than three. (The reader is encouraged to review the original for a reminder of the significance of Z scores in this context.) Scaramozzino et al. also mention employing community detection algorithms to achieve this result. It is also worth noting that this is a directed graph and yet only one cycle emerges—in pink, to the right of the center of the graph, including AVGO, JNJ, and PM.

Figure 1: Community Detection (Company Level)



Review this and the other six graphs (seven total) from the Scaramozzino et al. article with your new knowledge of network topology. Consider and explain the ways you can describe these graphs with the concepts from Coscia. Do Scaramozzino et al. deal with simple and/or complex networks? Do they discuss bi-partite graphs? Multi-layer graphs? If the latter, are they one-to-one or many-to-many? Do aspects come into play? Are any of the networks signed? Are any of the networks dynamic? Are there any hypergraphs?

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Note too that the practical issue of binning returns in this reading, along with a new approach to the problem that addresses the special issue that arises in degree distributions. Recall Scaramozzino et al.'s warning: "It is evident that the appropriate choice of bins is crucial." Both sets of authors agree that equally sized bins "with the same increment for the entire space" do not yield the most informative view of the structure of your data: "for low degree values you're putting together very populated bins, while for high degree values usually the distribution is so dispersed that you aren't actually grouping together anything" (Coscia 99). However, where Scaramozzino et al. chose equiprobable bin sizes (so that each bin has approximately the same number of data points), Coscia adds a new binning technique, "power binning":

"You start with a small bin size, usually equal to 1. Then each bin becomes progressively larger, by a constant multiplicative factor. At first, the bins are still small. But, as you progress, the bins start to be large enough to group a significant portion of your space" (Coscia 100).

Coscia also observes that powerbinning in combination with the CCDF, or complementary cumulative distribution function, can often be the most visually compelling view of your data's degree distribution (100). The CCDF "is a probability for a node to have a degree k or higher" (Coscia 100).

What follows are a few additional features of the CCDF (F below) that should make sense to you if you understand the concept well:

- $F(0)=1$
- $F(0)=1$
- $F(1)=0$
- $F(k_{\max}+1)=0$
- the function decreases monotonically along k

(Sayama 391)

4. Conclusion

In this lesson, we learned more vocabulary and theory related to networks to prepare us for a deeper dive into Bayesian networks in the following lessons of this module.

References

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